ALPHA SYSTEM REPLACEMENT

GDPS Windows File Cluster and Legacy SMB Namespace Standard

WSFC • File Share Witness • Legacy UNC Preservation • Cutover and Rehearsal

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System	GDPS
Document	Windows File Cluster and Legacy SMB Namespace Standard
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Establishes the GDPS clustered SMB service model, witness/quorum design, namespace preservation, test/rehearsal model, and ordered production cutover sequence.

Executive Summary

GDPS must replace the Alpha file-service infrastructure without changing the UNC interfaces consumed by existing legacy Windows applications. Those applications currently access files through server identities such as \\alpha0, \\alpha3, and \\alpha6 and cannot be modified as part of the hardware migration. The GDPS file-service architecture therefore preserves those names as clustered SMB client-access identities after production cutover.

TEST, PROD, DR, and COOP use two-node Windows Server Failover Clustering (WSFC) for the clustered file/service tier. HQ and IM are separate data centers. TEST and PROD are cross-data-center clusters with one physical node at HQ and the peer node at IM. Each two-node cluster uses a File Share Witness for quorum. The witness is independent of the clustered business-data storage and exists only to participate in quorum decisions.

Physical-node naming baseline: TEST = HQVMGDPSFST1 / IMVMGDPSFST2; PROD = HQVMGDPSFSP1 / IMVMGDPSFSP2; DR = IMVMGDPSFSD1 / IMVMGDPSFSD2; COOP = IMVMGDPSFSC1 / IMVMGDPSFSC2. These names are architecture-controlled and are not administrator-selected placeholders.

Cross-data-center baseline: HQ and IM are separate data centers. TEST is split between HQVMGDPSFST1 at HQ and IMVMGDPSFST2 at IM. PROD is split between HQVMGDPSFSP1 at HQ and IMVMGDPSFSP2 at IM. The cluster, storage/data path, witness, DNS/client-access identity, and network design must permit the surviving node/site to own the service after loss of either data center.

Before production cutover, the legacy production identities alpha0, alpha3, and alpha6 must not be configured, registered, advertised, resolvable to GDPS, or usable from the GDPS environment. GDPS build, migration, replication, validation, and rehearsal activities use temporary identities such as alpha0-1, alpha3-1, and alpha6-1.

TEST provides the production rehearsal capability Alpha does not have today. Because TEST mirrors the production topology, the team can exercise cluster failover, witness behavior, shared storage, replication, SMB shares, permissions, File Processor access, Middleware and Delivery Worker behavior, PostgreSQL dependencies, cutover sequencing, and recovery procedures without relying on the sandbox. The only production behavior that cannot be exercised exactly before cutover is ownership of the actual alpha0, alpha3, and alpha6 production identities.

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1. Purpose and Scope

This standard defines the Windows clustered file-service architecture and legacy SMB namespace behavior required for GDPS. It covers TEST, PROD, DR, and COOP file clusters, the File Share Witness quorum model, temporary migration identities, preservation of legacy UNC paths, production cutover sequencing, rollback, and the use of TEST as the full production-rehearsal environment.

The standard applies to the SMB/file-service boundary required by unchanged legacy consumers. It does not make SMB the internal communication mechanism for all GDPS components.

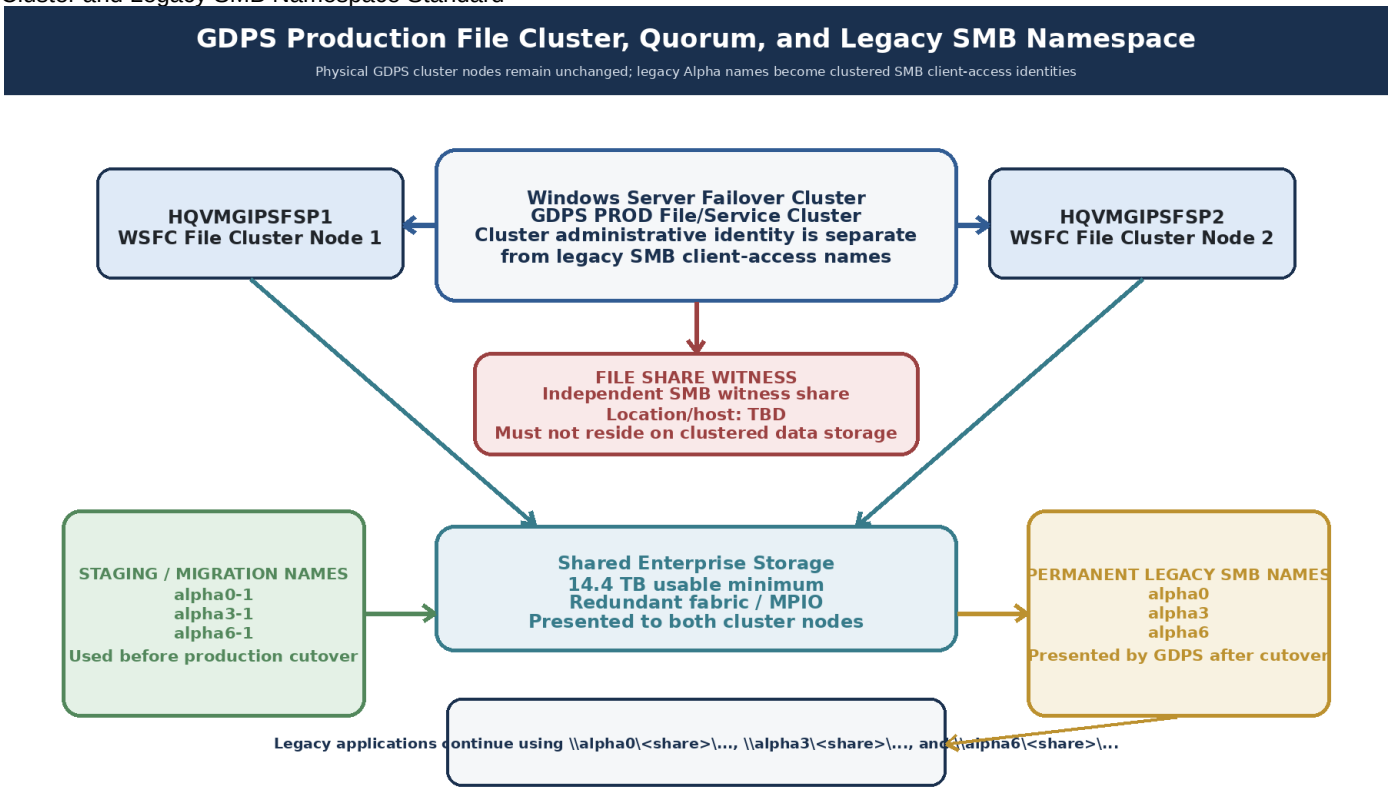
2. Governing Lift-and-Shift Principles

- Preserve the externally consumed Alpha SMB interfaces while replacing the unsupported Alpha/OpenVMS backend technology.
- Do not require legacy applications to change existing UNC references during this infrastructure migration.
- Preserve clustered availability, shared-storage behavior, replication, production failover, and COOP recovery semantics.
- Do not rename GDPS physical cluster nodes to legacy Alpha names.
- Treat alpha0, alpha3, and alpha6 as service identities that may be owned by only one production environment at a time.
- Do not configure or expose the final production Alpha SMB identities on GDPS before authorized cutover.
- Use TEST to rehearse production cluster, failover, replication, SMB, and cutover behavior before PROD.

3. GDPS Windows File Cluster Standard

Environment	Node 1	Node 2	Cluster Model	Storage	Purpose
TEST	HQVMGIPSFST1	HQVMGIPSFST2	2-node WSFC	Enterprise shared storage	Production mirror
PROD	HQVMGDPSFSP1	IMVMGDPSFSP2	2-node WSFC	Enterprise shared storage	Production clustered file/service tier
COOP	IMVMGIPSFSC1	IMVMGIPSFSC2	2-node WSFC	Enterprise crate storage	Recovery clustered file/service tier
DR	IMVMGDPSFSD1	IMVMGDPSFSD2	2-node WSFC	Enterprise shared recovery storage	Production Backup / DR recovery file/service tier

- The two physical nodes are independent computers participating in one Windows Server Failover Cluster.
- The cluster nodes use permanent GDPS infrastructure hostnames.
- Clustered SMB services are presented independently of whichever physical node currently owns the clustered role.
- The physical node hostname and the SMB client-access identity are separate concepts.
- Shared enterprise storage is presented to both file-cluster nodes through redundant paths.



GDPS PROD file cluster, File Share Witness, migration identities, and permanent legacy SMB client-access identities.

4. File Share Witness and Quorum Standard

Each two-node GDPS file cluster uses a File Share Witness as the quorum witness. The witness provides an additional vote so the cluster can make a deterministic ownership decision when only one cluster node remains available.

Environment	Witness Type	Witness Location	Requirement
TEST	File Share Witness	TBD independent witness host/share	Must not reside on TEST clustered data storage
PROD	File Share Witness	TBD independent witness host/share	Must not reside on PROD clustered data storage
COOP	File Share Witness	TBD COOP-reachable independent witness host/share	Must remain available during COOP operations and must not reside on crate clustered data storage
DR	File Share Witness	TBD independent witness host/share	Must not reside on DR clustered data storage; must support DR operation

- The witness share is quorum infrastructure only; it does not hold GDPS business data.
- The witness host must not depend on the clustered storage that it is helping arbitrate. For cross-data-center TEST and PROD, the witness must also remain reachable from the surviving site if either HQ or IM is unavailable.
- Witness permissions, cluster access, availability, monitoring, and backup of its configuration must be validated before the cluster is considered production-ready.

5. Legacy SMB Namespace Preservation

The existing Alpha SMB server names are part of the application interface. For example, in the path \\alpha3\<share>\<subfolder>, alpha3 is the SMB server/network identity, <share> is the SMB share name, and <subfolder> is the path below that share.

Because the consuming applications cannot be changed at this time, the production GDPS file cluster must ultimately present the same legacy SMB server identities and required share/path structure.

Name	Function	Ownership Rule
alpha0	Legacy production SMB client-access identity	Legacy Alpha only before cutover; GDPS only after cutover
alpha3	Legacy production SMB client-access identity	Legacy Alpha only before cutover; GDPS only after cutover
alpha6	Legacy production SMB client-access identity	Legacy Alpha only before cutover; GDPS only after cutover
HQVMGDPSFSP1	Physical GDPS PROD file-cluster node	Always GDPS
IMVMGDPSFSP2	Physical GDPS PROD file-cluster node	Always GDPS

Hard pre-cutover rule: alpha0, alpha3, and alpha6 must not be configured, registered, advertised, resolvable to GDPS, or usable on GDPS before the production cutover.

6. Temporary Migration Namespace

GDPS uses temporary SMB identities during build, migration, replication, testing, and production rehearsal. These identities allow the replacement environment to be fully exercised without creating a network/name conflict with the still-active legacy Alpha systems.

Temporary Name	Purpose	Permitted Use	Final State
alpha0-1	Temporary migration/staging identity for the alpha0 interface	Build, copy, validation, rehearsal	Removed or retired after production cutover
alpha3-1	Temporary migration/staging identity for the alpha3 interface	Build, copy, validation, rehearsal	Removed or retired after production cutover
alpha6-1	Temporary migration/staging identity for the alpha6 interface	Build, copy, validation, rehearsal	Removed or retired after production cutover

7. Share and Path Preservation

- Every required legacy SMB share name beneath alpha0, alpha3, and alpha6 must be inventoried before migration.
- The required directory/subdirectory structure beneath each share must be reproduced on GDPS.
- Share-level permissions and filesystem permissions must be mapped and tested.
- Service-account access required by legacy consumers must be verified through the temporary migration identities before cutover.
- The final GDPS production namespace must allow unchanged legacy applications to continue using their existing UNC paths.

Legacy Production Path	GDPS Pre-Cutover Validation Path	GDPS Final Production Path
\\alpha0\<share>\<subfolder>	\\alpha0-1\<share>\<subfolder>	\\alpha0\<share>\<subfolder>
\\alpha3\<share>\<subfolder>	\\alpha3-1\<share>\<subfolder>	\\alpha3\<share>\<subfolder>
\\alpha6\<share>\<subfolder>	\\alpha6-1\<share>\<subfolder>	\\alpha6\<share>\<subfolder>

8. TEST as the Production Rehearsal Environment

TEST is the deliberate improvement over the current Alpha environment. It is not a lightweight functional test environment. TEST mirrors the production infrastructure pattern so the team can validate the operational behavior that will exist in PROD before the production cutover.

- Two-node WSFC file/service cluster.
- File Share Witness quorum.
- Production-equivalent shared enterprise storage architecture.
- Same redundant storage-path, cross-data-center placement, node failover, and full-site failover mechanics as PROD.
- Same File Processor deployment pattern as PROD.
- Same Middleware, Issuer, and Delivery Worker deployment pattern as PROD.
- Same PostgreSQL high-availability pattern as PROD.
- Same service-account and permission model, using TEST identities and endpoints.
- Same replication and failover procedures, using TEST-specific data and names.
- Same software versions and deployment sequence intended for PROD.

9. End-to-End Validation Without Sandbox Dependency

The GDPS DEV and TEST environments provide sufficient coverage to exercise the replacement platform without relying on a separate sandbox for production-readiness validation. The sandbox may remain useful for isolated experiments, but it is not required to prove the production hardware, cluster, SMB, replication, or cutover design.

Validation Area	Environment	What Is Proven
Application build and developer testing	DEV	Normal development and component-level testing
External shared storage access	DEV	Validate UNC/NAS interaction using the DEV storage boundary
File Processor execution	DEV and TEST	Validate processing against configured shared storage
Middleware/API and Issuer	DEV and TEST	Validate API, authentication, and operational access
Delivery Worker	DEV and TEST	Validate downstream delivery behavior against nonproduction targets
WSFC cluster failover	TEST	Exercise production-equivalent cluster role movement

File Share Witness behavior	TEST	Exercise node loss and quorum behavior
Shared-storage failover / multipath	TEST	Validate storage path resilience
Replication mechanics	TEST	Exercise the same replication method intended for PROD
Share names, ACLs, service-account access	TEST	Validate complete file-service configuration with TEST or temporary names
Production cutover sequence	TEST	Rehearse every step except ownership of the real alpha0/alpha3/alpha6 identities
COOP recovery procedure	TEST + COOP exercises	Validate recovery sequencing and replicated-data readiness

Pre-cutover limitation: the only production behavior that cannot be exercised exactly in advance is activation of the real alpha0, alpha3, and alpha6 identities on GDPS, because those names remain exclusively owned by the legacy Alpha environment until cutover.

10. Production Cutover Approach — Order and Sequence

The production cutover is an identity-and-data transition, not simply a server startup. The sequence below prevents both the legacy Alpha environment and GDPS from simultaneously advertising the same production SMB identities.

Phase 1 — Build and Baseline

1. Build the GDPS PROD file cluster using only the permanent GDPS physical node names.
2. Configure shared storage, redundant paths, clustering, File Share Witness, monitoring, service accounts, and required network access.
3. Create and validate required shares through temporary identities alpha0-1, alpha3-1, and alpha6-1.
4. Confirm that alpha0, alpha3, and alpha6 are not configured or advertised by GDPS.

Phase 2 — Data Migration and Replication

5. Seed the GDPS production storage from the existing Alpha production data.
6. Establish the defined production-to-GDPS replication/synchronization mechanism.
7. Verify that required share and folder structures are reproduced.
8. Validate file counts, sizes, timestamps, and checksums or other agreed integrity evidence.

Phase 3 — Pre-Cutover Rehearsal

9. Execute the complete cutover procedure in TEST using TEST-specific or temporary names, including a planned data-center-loss rehearsal.
10. Exercise WSFC node failover, complete HQ-site loss, complete IM-site loss, File Share Witness behavior, storage/data failover, replication interruption/recovery, share access, and application access.
11. Validate rollback sequencing in TEST.
12. Record test evidence and unresolved defects. Production cutover cannot proceed with unresolved critical defects.

Phase 4 — Production Change Freeze

13. Announce and enter the approved production change window.
14. Stop or suspend Alpha processing that can create new files or modify files being migrated.
15. Stop new writes from upstream/downstream processes as required by the cutover runbook.

16. Confirm active file sessions/locks and incomplete jobs have been resolved.
17. Record the final source-state checkpoint.

Phase 5 — Final Synchronization

18. Run the final synchronization from Alpha production to GDPS production storage.
19. Confirm synchronization has completed successfully.
20. Perform final data validation against the source checkpoint.
21. Do not proceed to namespace transfer if the final data validation fails.

Phase 6 — Legacy Identity Shutdown

22. Disable the legacy Alpha SMB service/network identities alpha0, alpha3, and alpha6.
23. Remove or disable the legacy name registrations that would allow those identities to continue resolving to the old environment.
24. Confirm that the old systems no longer answer SMB requests for alpha0, alpha3, or alpha6.
25. Confirm there is no duplicate production identity visible on the network.

Phase 7 — GDPS Production Identity Activation

26. Activate alpha0, alpha3, and alpha6 as clustered SMB client-access identities on the GDPS production file cluster.
27. Establish the required DNS/name registration and authentication bindings for the new clustered identities.
28. Expose the required legacy share names beneath the correct identities.
29. Confirm that the physical node names remain HQVMGDPSFSP1 and IMVMGDPSFSP2.

Phase 8 — Production Validation

30. Validate access to representative paths under \\alpha0, \\alpha3, and \\alpha6.
31. Validate share permissions, filesystem permissions, and service-account access.
32. Validate representative unchanged legacy applications without modifying their configured UNC paths.
33. Validate File Processor, Middleware/Delivery Worker, and other GDPS services that depend on the file-service boundary.
34. Perform a controlled cluster failover test if the production change plan authorizes it during the validation window.

Phase 9 — Release and Stabilization

35. Release GDPS production processing.
36. Resume upstream and downstream integrations in the approved order.
37. Monitor WSFC health, witness health, storage paths, SMB sessions, replication state, application errors, and processing throughput.
38. Retain the temporary migration identities only as long as the approved stabilization plan requires; then remove them.

11. Cutover Entry Criteria

- TEST production-mirror rehearsal completed successfully.
- All required PROD cluster nodes, witness, shared storage/data-replication capability, MPIO/redundant paths, monitoring, and network dependencies are operational across both HQ and IM, and TEST has successfully demonstrated service continuity after simulated loss of either data center.
- Temporary migration identities and all required share/path structures have been validated.
- Replication/synchronization process has been proven and the latest replication state is healthy.

- Legacy application representatives have validated required nonproduction share access where practical.
- Production cutover window, ownership, communication plan, and rollback authority are established.
- Backout/rollback decision point is explicitly identified before the cutover begins.

12. Cutover Stop Conditions

- Final synchronization does not complete successfully.
- File counts, checksums, required timestamps, or other agreed data-integrity evidence do not reconcile.
- alpha0, alpha3, or alpha6 is still active on the legacy environment when GDPS activation is about to begin.
- A duplicate SMB/DNS identity is detected.
- WSFC quorum or File Share Witness is unhealthy.
- Shared-storage or MPIO path health is degraded beyond the accepted cutover threshold.
- Required share names or permissions are missing.
- Representative legacy application access fails after identity activation and cannot be corrected within the approved validation window.

13. Rollback Approach

Rollback must distinguish between a failure that occurs before GDPS accepts authoritative production writes and a failure that occurs after new GDPS writes have begun.

Rollback Point	Required Action	Rule
Before GDPS authoritative writes begin	Disable GDPS production SMB identities; restore the legacy Alpha identities; verify legacy SMB access; resume Alpha processing after source state is confirmed.	Straight namespace rollback is permitted because no new authoritative GDPS writes must be reconciled.
After GDPS authoritative writes begin	Stop further GDPS writes; preserve both source and GDPS state; perform controlled data reconciliation; only restore Alpha identities after the authoritative data set has been determined.	Do not simply switch names back. Data reconciliation is mandatory.

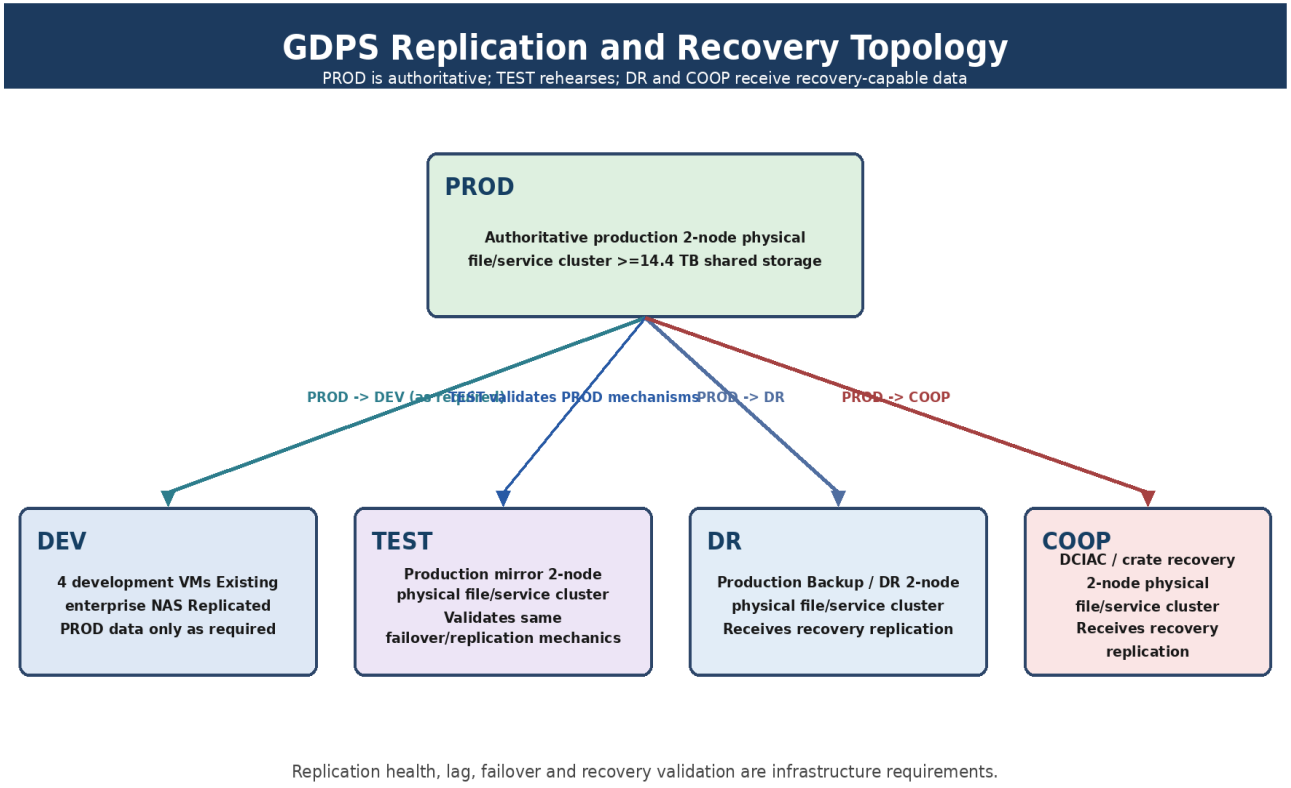
14. Post-Cutover Validation and Stabilization

- Confirm alpha0, alpha3, and alpha6 resolve only to the GDPS clustered service.
- Confirm all required SMB shares are online and owned by the appropriate clustered role.
- Confirm representative legacy applications operate with no UNC-path changes.
- Confirm service accounts and scheduled/automated jobs can read/write required locations.
- Confirm File Processor access to configured shared folders.
- Confirm Delivery Worker delivery to preserved SMB destinations where required.
- Confirm replication from PROD to COOP is operating and within the accepted lag threshold.
- Confirm cluster events, witness state, storage paths, and SMB errors are being monitored.
- Capture the final production validation evidence and cutover completion record.

15. COOP Namespace and Recovery Considerations

DR and COOP are recovery environments, not simultaneously active owners of the production SMB names. DR replaces/improves the ALPHA6 Production Backup capability, while COOP preserves the DCIAC recovery capability. During normal production operation, the real production names alpha0, alpha3, and alpha6 remain owned by PROD.

- COOP maintains its own two-node file cluster, File Share Witness, crate storage, and replicated production data.
- COOP testing uses COOP-specific or temporary names; it does not advertise the real production identities during normal operations.
- During a declared PROD-to-COOP recovery, the production SMB identities may be transferred to COOP only after PROD is confirmed unable to advertise or serve those identities.
- The same single-owner rule applies during recovery: only one environment may advertise alpha0, alpha3, or alpha6 at a time.
- Return from COOP to PROD requires the same controlled data synchronization, identity release, identity activation, validation, and rollback discipline as the original production cutover.



GDPS replication, DR, COOP, and TEST recovery topology.

16. Operational Ownership and Evidence

Owner	Responsibility	Required Evidence
Infrastructure / Windows Cluster	WSFC configuration, cluster nodes, quorum/witness, clustered roles, node health	Cluster validation report; failover results
Storage Engineering	Shared storage, redundant paths, multipath configuration, replication platform	Storage health report; path validation; replication status
Network / DNS / AD	Name registration, DNS, SMB identity support, authentication/SPN configuration as required	Name-resolution and authentication validation
GDPS Application Team	Share/path requirements, File Processor/Middleware/Delivery validation, application access testing	Application and processing validation evidence
Legacy Application Owners	Representative unchanged application testing against required UNC interfaces	Consumer acceptance evidence
Cutover Lead	Sequence control, stop/go decisions, rollback authority, communications	Approved cutover record and completion checklist

17. Implementation Guardrails

- alpha0, alpha3, and alpha6 must never be usable from GDPS before production cutover.
- Do not allow Alpha and GDPS to advertise the same production SMB client-access identity simultaneously.
- Do not rename HQVMGDPSFSP1 or IMVMGDPSFSP2 to an Alpha name.
- Do not use the File Share Witness as business-data storage.
- Do not place the witness on the same clustered storage that it is arbitrating.
- Do not replace clustered shared storage with local application-server disks.
- Do not use TEST as a reduced environment; TEST must remain a production mirror for infrastructure behavior, including cross-data-center HQ/IM placement and site-loss failover.
- Do not depend on a sandbox to prove production readiness when DEV and TEST already provide the required validation path.
- Do not change legacy UNC paths during this infrastructure cutover unless a separately approved application change is introduced.
- Do not treat SMB as the internal transport for all GDPS services; SMB is retained where file-based compatibility and shared-file access require it.